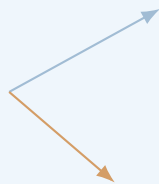


Physics Experiments

General Notes

Physics Experiments

Independent physical ideas, models, and exploratory notes.



June 6, 2026

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Chapter 1

Template Independent Note

1.1 Seed Idea

Write the core idea here. This file is a placeholder for an independent physics note created from a discussion, a loose thought, or a larger experiment.

1.2 Working Notes

Use this section to develop physical interpretation, assumptions, models, equations, diagrams, examples, and open questions. The goal is not to force the idea into a course sequence, but to make the thought readable enough that it can later be revised, expanded, or moved into a formal lecture-note project.

Remark

This chapter is intentionally independent. Future notes should usually be added as new `chapter_*.tex` files rather than inserted into this template.

1.3 Possible Next Steps

- Identify the physical system and assumptions.
- Add units, dimensional checks, and limiting cases.
- Record any useful diagrams in `figures/independent_notes/`.
- Decide later whether the note belongs in a formal course project.

Chapter 2

Projectile Motion as a Two-Dimensional Model

2.1 Purpose and Physical Assumptions

Projectile motion is one of the simplest situations in which a two-dimensional motion becomes understandable by separating it into two one-dimensional motions. The horizontal motion is uniform, while the vertical motion is uniformly accelerated by gravity.

In this note we derive the standard formulas for an ideal projectile launched from a point with initial speed v_0 at an angle θ above the horizontal. The purpose is not only to list the formulas, but to show where they come from and what assumptions make them valid.

We use the following idealized model.

- The projectile is treated as a point particle.
- Air resistance is neglected.
- The gravitational acceleration is constant and directed downward.
- The Earth is treated locally as flat, so horizontal and vertical axes are meaningful over the distance considered.
- The launch point is chosen as the origin unless stated otherwise.

With the positive y -axis pointing upward, the acceleration vector is

$$\mathbf{a} = (0, -g),$$

where $g > 0$ is the magnitude of gravitational acceleration.

2.2 Geometry of the Launch

Let the initial velocity be

$$\mathbf{v}_0 = (v_{0x}, v_{0y}).$$

If the initial speed is $v_0 = \|\mathbf{v}_0\|$ and the launch angle is θ , then the velocity components are

$$v_{0x} = v_0 \cos \theta, \quad v_{0y} = v_0 \sin \theta.$$

This is simply the decomposition of a vector into its horizontal and vertical components.

2.3 Equations of Motion from Constant Acceleration

The acceleration equations are

$$\frac{d^2x}{dt^2} = 0, \quad \frac{d^2y}{dt^2} = -g.$$

We derive the position functions by integrating with respect to time.

For the horizontal coordinate,

$$\frac{d^2x}{dt^2} = 0.$$

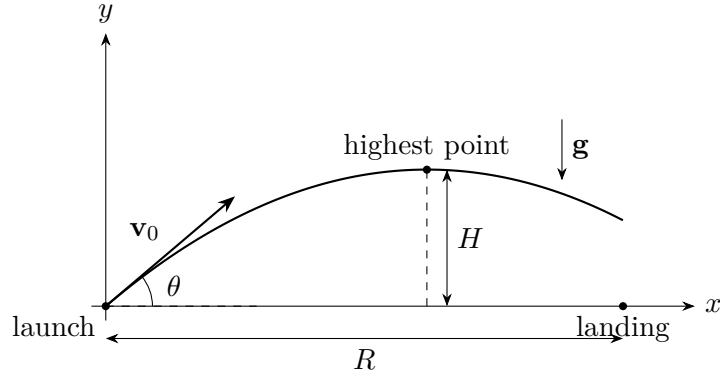


Figure 2.1: Ideal projectile motion without air resistance. The trajectory is a parabola, the acceleration is vertical, and the horizontal and vertical motions can be treated separately.

Integrating once gives

$$\frac{dx}{dt} = C_1.$$

The constant C_1 is the initial horizontal velocity, so

$$\frac{dx}{dt} = v_{0x} = v_0 \cos \theta.$$

Integrating again gives

$$x(t) = v_{0x}t + C_2.$$

If $x(0) = 0$, then $C_2 = 0$, hence

$$\boxed{x(t) = v_0 \cos \theta t.}$$

For the vertical coordinate,

$$\frac{d^2y}{dt^2} = -g.$$

Integrating once gives

$$\frac{dy}{dt} = -gt + C_3.$$

The constant C_3 is the initial vertical velocity, so

$$\frac{dy}{dt} = v_{0y} - gt = v_0 \sin \theta - gt.$$

Integrating again gives

$$y(t) = v_{0y}t - \frac{1}{2}gt^2 + C_4.$$

If $y(0) = 0$, then $C_4 = 0$, hence

$$\boxed{y(t) = v_0 \sin \theta t - \frac{1}{2}gt^2.}$$

Thus the velocity and position are

$$\boxed{\mathbf{v}(t) = (v_0 \cos \theta, v_0 \sin \theta - gt)}$$

and

$$\boxed{\mathbf{r}(t) = (v_0 \cos \theta t, v_0 \sin \theta t - \frac{1}{2}gt^2).}$$

Remark

The horizontal coordinate changes linearly in time because there is no horizontal acceleration in the ideal model. The vertical coordinate contains a quadratic term because the vertical acceleration is constant and nonzero.

2.4 The Trajectory as a Parabola

The parametric equations are

$$x(t) = v_0 \cos \theta t, \quad y(t) = v_0 \sin \theta t - \frac{1}{2}gt^2.$$

Assume $\cos \theta \neq 0$. From the horizontal equation,

$$t = \frac{x}{v_0 \cos \theta}.$$

Substituting this into $y(t)$ gives

$$y = x \tan \theta - \frac{g}{2v_0^2 \cos^2 \theta} x^2.$$

Therefore the trajectory is

$$y(x) = x \tan \theta - \frac{g}{2v_0^2 \cos^2 \theta} x^2.$$

This is a quadratic function of x , so the ideal trajectory is a parabola.

2.5 Time of Flight

The projectile returns to the launch height when $y(t) = 0$. With launch and landing at the same height,

$$y(t) = v_0 \sin \theta t - \frac{1}{2}gt^2 = 0.$$

Factoring out t ,

$$t \left(v_0 \sin \theta - \frac{1}{2}gt \right) = 0.$$

There are two solutions. The first is $t = 0$, the launch moment. The second is found from

$$v_0 \sin \theta - \frac{1}{2}gt = 0.$$

Hence the total time of flight is

$$T = \frac{2v_0 \sin \theta}{g}.$$

This formula applies when the projectile lands at the same vertical level from which it was launched.

2.6 Maximum Height

At the highest point the vertical velocity is zero:

$$v_y(t) = v_0 \sin \theta - gt = 0.$$

Therefore the time to reach the top is

$$t_H = \frac{v_0 \sin \theta}{g}.$$

Substituting t_H into $y(t)$,

$$H = v_0 \sin \theta \left(\frac{v_0 \sin \theta}{g} \right) - \frac{1}{2} g \left(\frac{v_0 \sin \theta}{g} \right)^2.$$

Thus

$$H = \frac{v_0^2 \sin^2 \theta}{g} - \frac{v_0^2 \sin^2 \theta}{2g} = \frac{v_0^2 \sin^2 \theta}{2g}.$$

So the maximum height above the launch point is

$$H = \frac{v_0^2 \sin^2 \theta}{2g}.$$

2.7 Horizontal Range

The range R is the horizontal distance traveled during the total time of flight. Since

$$x(t) = v_0 \cos \theta t,$$

we have

$$R = x(T) = v_0 \cos \theta T.$$

Using $T = 2v_0 \sin \theta / g$,

$$R = v_0 \cos \theta \cdot \frac{2v_0 \sin \theta}{g} = \frac{2v_0^2 \sin \theta \cos \theta}{g}.$$

Since $2 \sin \theta \cos \theta = \sin(2\theta)$,

$$R = \frac{v_0^2 \sin(2\theta)}{g}.$$

The range is maximal when $\sin(2\theta) = 1$, that is when

$$2\theta = 90^\circ, \quad \theta = 45^\circ.$$

For launch and landing at the same height, the ideal range is therefore maximal at 45° .

2.8 Velocity, Speed, and Direction During Flight

The velocity components are

$$v_x(t) = v_0 \cos \theta, \quad v_y(t) = v_0 \sin \theta - gt.$$

The speed is the magnitude of the velocity vector:

$$\|\mathbf{v}(t)\| = \sqrt{v_x(t)^2 + v_y(t)^2} = \sqrt{v_0^2 \cos^2 \theta + (v_0 \sin \theta - gt)^2}.$$

Thus

$$\|\mathbf{v}(t)\| = \sqrt{v_0^2 \cos^2 \theta + (v_0 \sin \theta - gt)^2}.$$

The instantaneous direction angle $\alpha(t)$ of the velocity satisfies

$$\tan \alpha(t) = \frac{v_y(t)}{v_x(t)} = \frac{v_0 \sin \theta - gt}{v_0 \cos \theta},$$

provided $v_0 \cos \theta \neq 0$.

At the highest point, $v_y = 0$, so the velocity is purely horizontal:

$$\mathbf{v}(t_H) = (v_0 \cos \theta, 0).$$

2.9 Projectile Launched from an Initial Height

If the projectile is launched from height y_0 , the position equations become

$$x(t) = x_0 + v_0 \cos \theta t,$$

and

$$y(t) = y_0 + v_0 \sin \theta t - \frac{1}{2}gt^2.$$

If the ground is at $y = 0$, the landing time is obtained from

$$y_0 + v_0 \sin \theta t - \frac{1}{2}gt^2 = 0.$$

Equivalently,

$$gt^2 - 2v_0 \sin \theta t - 2y_0 = 0.$$

The physically relevant positive solution is

$$T = \frac{v_0 \sin \theta + \sqrt{v_0^2 \sin^2 \theta + 2gy_0}}{g}.$$

The corresponding range, measured from x_0 , is

$$R = v_0 \cos \theta T.$$

When $y_0 = 0$, the formula reduces to

$$T = \frac{v_0 \sin \theta + |v_0 \sin \theta|}{g}.$$

For an upward launch with $\sin \theta > 0$, this becomes

$$T = \frac{2v_0 \sin \theta}{g},$$

which agrees with the same-height result.

2.10 Energy Check

The same maximum height formula can be checked using conservation of mechanical energy. At launch, the vertical part of the kinetic energy associated with upward motion is

$$\frac{1}{2}m(v_0 \sin \theta)^2.$$

At the highest point, this vertical kinetic energy has been converted into gravitational potential energy mgH . Thus

$$\frac{1}{2}mv_0^2 \sin^2 \theta = mgH.$$

Canceling m and solving for H ,

$$H = \frac{v_0^2 \sin^2 \theta}{2g}.$$

This agrees with the kinematic derivation.

2.11 A Worked Symbolic Example

Suppose a projectile is launched from ground level with speed $v_0 = 20 \text{ m/s}$ at angle $\theta = 30^\circ$. Since

$$\sin 30^\circ = \frac{1}{2}, \quad \cos 30^\circ = \frac{\sqrt{3}}{2},$$

the initial components are

$$v_{0x} = 10\sqrt{3} \text{ m/s}, \quad v_{0y} = 10 \text{ m/s}.$$

The time of flight is

$$T = \frac{2v_{0y}}{g} = \frac{20}{g}.$$

The maximum height is

$$H = \frac{v_{0y}^2}{2g} = \frac{100}{2g} = \frac{50}{g}.$$

The range is

$$R = v_{0x}T = 10\sqrt{3} \cdot \frac{20}{g} = \frac{200\sqrt{3}}{g}.$$

If one later substitutes $g \approx 9.81 \text{ m/s}^2$, these symbolic expressions give numerical predictions in SI units.

2.12 Limitations of the Model

The ideal projectile model is powerful because it is simple, but its assumptions are restrictive. In real motion, air resistance may reduce the range, change the optimal launch angle, and make the trajectory non-parabolic. For high speeds or long distances, the curvature of the Earth and variation of g may also matter. The formulas derived here are therefore best understood as exact results inside an ideal model, and as approximations for real projectiles when the assumptions are sufficiently accurate.

Summary

Projectile motion without air resistance separates into horizontal uniform motion and vertical uniformly accelerated motion. From $\mathbf{a} = (0, -g)$, integration gives

$$x(t) = v_0 \cos \theta t, \quad y(t) = v_0 \sin \theta t - \frac{1}{2}gt^2.$$

Eliminating time gives a parabolic trajectory. For launch and landing at the same height,

$$T = \frac{2v_0 \sin \theta}{g}, \quad H = \frac{v_0^2 \sin^2 \theta}{2g}, \quad R = \frac{v_0^2 \sin(2\theta)}{g}.$$